

BRAC University
Department of Computer Science and Engineering
CSE420:Compiler Design

Examination: Quiz 4
Time: 25 min

Semester: Fall 25
Marks: 15

Name:	ID:	Section:
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1. Consider the following SDD 1 and SDD 2. Draw the Annotated Parse Tree and Evaluate it for the String: **if (a < b) a = - b ; else b = - a ;** Also, generate the Three Address Code (3AC).

PRODUCTION	SEMANTIC RULES
$P \rightarrow S$	$S.next = newlabel()$ $P.code = S.code \parallel label(S.next)$
$S \rightarrow \text{if } (B) S_1$	$B.true = newlabel()$ $B.false = S_1.next = S.next$ $S.code = B.code \parallel label(B.true) \parallel S_1.code$
$S \rightarrow \text{if } (B) S_1 \text{ else } S_2$	$B.true = newlabel()$ $B.false = newlabel()$ $S_1.next = S_2.next = S.next$ $S.code = B.code$ $\parallel label(B.true) \parallel S_1.code$ $\parallel gen('goto' S.next)$ $\parallel label(B.false) \parallel S_2.code$
$S \rightarrow \text{while } (B) S_1$	$begin = newlabel()$ $B.true = newlabel()$ $B.false = S.next$ $S_1.next = begin$ $S.code = label(begin) \parallel B.code$ $\parallel label(B.true) \parallel S_1.code$ $\parallel gen('goto' begin)$
$S \rightarrow S_1 S_2$	$S_1.next = newlabel()$ $S_2.next = S.next$ $S.code = S_1.code \parallel label(S_1.next) \parallel S_2.code$

fig: SDD 1

$B \rightarrow E_1 \text{ rel } E_2$	$B.code = E_1.code \parallel E_2.code$ $\parallel \text{gen('if' } E_1.addr \text{ rel.op } E_2.addr \text{ 'goto' } B.true)$ $\parallel \text{gen('goto' } B.false)$
$S \rightarrow \text{id} = E ;$	$S.code = E.code \parallel$ $\text{gen}(top.get(\text{id.lexeme}) '=' E.addr)$
$E \rightarrow E_1 + E_2$	$E.addr = \text{new Temp}()$ $E.code = E_1.code \parallel E_2.code \parallel$ $\text{gen}(E.addr '=' E_1.addr '+' E_2.addr)$
$ - E_1$	$E.addr = \text{new Temp}()$ $E.code = E_1.code \parallel$ $\text{gen}(E.addr '=' \text{'minus' } E_1.addr)$
$ (E_1)$	$E.addr = E_1.addr$ $E.code = E_1.code$
$ \text{id}$	$E.addr = top.get(\text{id.lexeme})$ $E.code = ''$

fig: SDD 2

- At any point during program execution, the number of activation records on the run time stack is equal to the depth of the currently executing node in the activation tree.
State whether the statement is True or False [1] and justify your answer within 2–3 lines.
[4]